

Chapter 1

Grammatical Tone in Igbo Verbs

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This paper analyzes the understudied grammatical tone system in Igbo verbal paradigms, highlighting a distinction between two types of TAM morphemes: (i) those without floating tones, which do not override the verb root's underlying tone, and (ii) those with floating tones, which do. The results of our novel analysis demonstrate that floating high grammatical tones trigger epenthesis of an extra Tone Bearing Unit (a mora), while floating low grammatical tones override the verb root's underlying tone. The study offers a novel perspective on Igbo tone and suggests directions for future research.¹

1 Introduction

This paper investigates a previously seldom-discussed phenomenon in Igbo verbal morphology—grammatical tone. Igbo verbs exhibit various tense-aspect-mood (TAM) distinctions, marked by verbal clitics/affixes and tone changes. We analyze these morphemes as they appear with monosyllabic verbs, which constitute the vast majority of Igbo verbs (Emenanjo 1978; Zsiga 1992; Mbah 1999). While we intend to extend our analysis to multisyllabic and compound verbs in the future, we restrict our focus to monosyllabic verbs and exclude negation constructions in the present paper for clarity and space. Our findings indicate two

¹The authors are listed in alphabetical order.

distinct types of TAM morphemes. The first are preverbal morphemes², which have their own underlying tone and do not alter the verb root. The second are affixes containing segmental and/or tonal material, which do change the verb root's underlying tone.

These tonal affixes fall into two groups: those with a floating low (L) tone and those with a floating high (H) tone. Floating L tone overrides the verb's underlying tone, while floating H tone triggers epenthesis of an additional mora, creating a tone bearing unit (TBU) to realize the tone without overriding the underlying tone of the verb root.

These patterns have broader implications for Igbo morphophonology and morphophonological theory. Notably, floating grammatical tones in Igbo behave differently depending on whether the floating tone is H or L, suggesting that the phonology of floating grammatical tones is not uniform even within a single language. The fact that L tone overrides the root, while H tone triggers epenthesis, challenges prior assumptions that H is the default or active tone in Igbo (Clark 1990). We argue that this assumption should be reconsidered. As discussed in Section 5, these findings have implications for other areas of Igbo grammar. Section 6 also suggests that different tonal processes operate at the level of the prosodic word versus the prosodic phrase.

The remainder of this paper proceeds as follows. Section 2 provides background on Igbo phonology and morphosyntax. Section 3 presents the data for H and L tone monosyllabic verbs. Section 4 offers our analysis. Section 5 discusses the broader implications, and Section 6 concludes with future directions.

2 Background

2.1 The Igbo Language

The Igbo language is a West Benue-Congo language spoken primarily in the Southeastern part of Nigeria and Equatorial Guinea (Armstrong 1967). It is one of Nigeria's major languages, with an estimated eighteen million speakers (Nkamigbo 2014). It has about thirty varieties or dialects. The data in this paper comes from the Orlu dialect, a variety of Igbo spoken in the Imo-west geographical zone of Imo state, in South-eastern Nigeria, of which the first author is a native speaker.

²We remain agnostic as to whether these are free standing morphemes, prefixes, or proclitics, though they phonologically integrate with the verb.

Studies in Igbo phonology reveal that there are roughly eight phonemic oral vowels, divided into two sets based on the [$\pm$ ATR] features (Ikekeonwu 1999), and twenty-eight phonemic consonants (Nweke 2012).

2.2 The Phonological Inventory of Igbo

The phonemic inventory of the standard dialect of the Igbo comprises thirty-six phonemes. These include twenty-eight consonants (Table 1) and eight vowels (Figure 1).

Table 1: The Consonant Inventory of Igbo

	Labial	Lab-dent	Alveolar	Post-alv	Palatal	Velar	Lab-velar	Glottal
Stop	p b		t d			k g	$\widehat{kp} \widehat{gb}$ $k^w g^w$	
Nas	m		n		η	η	η^w	
Fric		f v	s z	$\int$		γ		h
Aff				$\widehat{t\int} \widehat{d\zeta}$				
Lat			l					
Trill			r					
Appr					j		w	

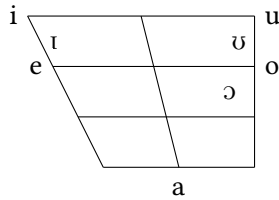


Figure 1: The Vowel Inventory of Igbo

2.2.1 ATR Distinction and Vowel Harmony

As previously noted, Igbo has eight phonemic vowels organized into two harmonic sets distinguished by the Advanced Tongue Root feature (ATR). Vowels in the [+ATR] set are produced with an advanced (forward) tongue root position,

while those in the [−ATR] set involve a retracted tongue root. Example (1) shows the [±ATR] vowel pairs.

- (1) [+ATR]: i, e, o, u
[−ATR]: ɪ, ʌ, ɔ, ʊ

2.2.2 The Tone System

Igbo exhibits a register tone system with two basic contrastive level tones: H and L (Clark 1990). There is also the presence of a Mid-tone, which is not considered a level tone and is more accurately described as a downstepped H (!H). There are also two contour tones, rising (LH) and falling (HL), that are not lexically distinctive; rather, they are derived by phonological rule application (Ugorji 2008).

The level tones in Igbo distinguish otherwise phonemically identical words. These distinctions are most common in nouns and are not lexically contrastive for verbs, though verbs do possess their own underlying tone. Example (2) illustrates lexical tonal contrasts in Igbo.

- | | | | |
|-----|-------------------|--|-------------------|
| (2) | a. HH | | c. LL |
| | ák ^w á | | àk ^w à |
| | ‘cry’ | | ‘bed’ |
| | b. HL | | d. LH |
| | ák ^w à | | àk ^w á |
| | ‘cloth’ | | ‘egg’ |

2.3 The Basic Morphosyntax of Igbo Verbal Constructions

Igbo has a canonical strict Subject-Verb-Object (SVO) word order. The verb root typically carries essential grammatical information and is inflected through tonal alternation, reduplication, and the addition of preverbal morphemes as well as suffixes that express tense, aspect, mood, and polarity. Object pronouns and noun complements, and adverbial phrases follow the verb directly. Verbs may also co-occur with auxiliary verbs, in serial verb constructions, and with inherent complements, which function to contribute to a rich and flexible Igbo verbal system. A typical basic clause illustrating the Igbo canonical word order is shown in (3).

- (3) *ézè é-tòó-n'á àdá*
 Eze v-praise-COMPL Ada
 'Eze (had) praised Ada'

Now, let us look at the different TAM alternations in Igbo and how they demonstrate grammatical tone.

3 Presentation of the Data

In this section, we outline the six TAM combinations for monosyllabic verbs that we will provide an analysis for in [Section 4](#). We begin with the progressive ([Section 3.1](#)), then the future ([Section 3.2](#)), the infinitive ([Section 3.3](#)), the perfective ([Section 3.4](#)), the completive ([Section 3.5](#)), and finally the imperative ([Section 3.6](#)).

3.1 The Progressive

The progressive aspect in Igbo is morphosyntactically marked by the preverbal morpheme /*nà-*/, which bears an L tone (4). The verb form in this construction typically exhibits a V.CV syllable structure, where the initial V-slot hosts a verbal prefix, and the following CV sequence corresponds to the verb stem ([Emenanjo 1978](#); [Clark 1990](#); [Zsiga 1992](#); [Mbah 1999](#)). Importantly, the progressive aspect in Igbo can only be used in the present tense, thus indicating an obligatory co-occurrence of present tense and progressive aspect ([Emenanjo 1985; 2016](#); [Obiamalu 2016](#)).

- | | | | | |
|-----|----|--|----|--|
| (4) | a. | <i>ézè nà-é-tò àdá</i>
Eze PROG-V-praise Ada
'Eze is praising Ada' | b. | <i>ézè nà-è-ḡbú àgù</i>
Eze PROG-V-kill lion
'Eze is killing (a/the) lion' |
|-----|----|--|----|--|

In our analysis, the initial vowel is analyzed as an abstract verbal prefix *v-*, which is realized phonologically as [e]- or [a]-, conditioned by the [$\pm$ ATR] feature of the verb root. The choice between [e]- and [a]- reflects vowel harmony with the verb stem ([+ATR] and [-ATR] respectively). Tonally, the prefix is subject to the Obligatory Contour Principle (OCP) such that if the stem vowel is underlyingly H, the prefix surfaces with L to avoid a sequence of adjacent H tones; conversely, if the root vowel is L, the prefix is H. Furthermore, we do not offer a gloss for this

vowel as it is unclear to us at the present time what its morphosyntactic function is.

3.2 The Future

The future tense in Igbo is marked by a preverbal morpheme /gà-/ bearing a L tone, and the verb stem takes the same V.CV form as the progressive. An illustrative example is given in (5).

- (5) a. *éze gà-é-tò àdá* b. *éze gà-è-gbú àgù*
 Eze FUT-V-praise Ada Eze FUT-V-kill lion
 ‘Eze will praise Ada’ ‘Eze will kill (a/the) lion’

As seen in (5), the future tense exhibits the same pattern for syllable structure and tone as the progressive.

3.3 The Infinitive

The infinitive in Igbo marks non-finite constructions and is realized with a H-toned prefix /í-/ or /í-/ to maintain vowel harmony with the verb root (6).

- (6) a. *í-tò àdá* b. *í-gb'ú àgù*
 INF-praise Ada INF-kill lion
 ‘to praise Ada’ ‘to kill (a/the) lion’

Unlike preverbal *v-*, the high tone on the infinitival prefix is invariant and does not undergo tonal alternation as a result of the OCP. Instead, the infinitive marker causes a H tone on the following verb root to become downstepped.

3.4 The Perfect

In Igbo, the perfective aspect is morphologically marked by a suffixal morpheme *-rV̌*, where the consonant *-r* is fixed and the *V̌* is a copy of the final vowel of the verb root. The entire *-rV̌* suffix consistently bears a L tone, regardless of the lexical tone of the verb (7).

- (7) a. *éze tò-rò àdá* *àdá*
 Eze praise-PRFV Ada
 ‘Eze (has) praised Ada’
- b. *éze gbù-rù àgù*
 Eze kill-PRFV lion
 ‘Eze (has) killed (a/the) lion’

Importantly, this L tone suffix also overrides the underlying tone of the verb root.

3.5 The Completive

The completive aspect in Igbo is realized through the same V.CV verb-stem as the progressive and future, with the addition of a suffix */-n'á/*. Like in other V.CV stems, the first vowel is the *v-* prefix that is selected by vowel harmony and OCP. Interestingly, this prefix only appears in constructions with lexical subject NPs (8). It does not surface in completive constructions with subject pronouns (9).

- (8) a. *éze é-tòó-n'á àdá* *àdá*
 Eze v-praise-COMPL Ada
 ‘Eze (had) praised Ada’
- b. *éze è-gbù'ó-n'á àgù*
 Eze v-kill-COMPL lion
 ‘Eze (had) killed (a/the) lion’
- (9) a. *ó tòó-n'á àdá*
 3.SG praise-COMPL Ada
 ‘he/she (had) praised Ada’
- b. *ó gbù'ó-n'á àgù*
 3.SG kill-COMPL lion
 ‘he/she (had) killed (a/the) lion’

A notable phonological feature of the completive is the insertion of an epenthetic mora following the final vowel of the verb root. When the root ends in a high vowel, a mid vowel is inserted to host the grammatical tone (e.g., *gbù* becomes *gbù'ó*); for non-high vowels, vowel lengthening occurs instead (e.g., *tò* becomes *tòó*). We propose that this moraic augmentation serves a prosodic function by providing a suitable TBU for an unassociated floating grammatical H tone at the left edge of the completive suffix (i.e., */-^Hn'á/*). This additional floating grammatical H tone also explains the downstepped H on the suffix */-n'á/* (Clark 1990).

3.6 The Imperative

In Igbo, the imperative is formed by overriding the verb root’s underlying tone with a LH melody (Clark 1990; Ikegwuonu 2014) and the subsequent lengthening

of the verb root's final vowel (10).

- (10) a. *tòó àdá*
praise Ada
'praise Ada!'
- b. *ḡbùò àgù*
kill lion
kill the lion!'

As we will argue in the next sections, we believe that the change in syllable structure is a prosodic fix in order to avoid having a contour tone on a single mora in Igbo. Thus, we will argue that the imperative is solely expounded by the LH melody.

4 Analysis

In the previous section (Section 3), it was demonstrated how the tone on a verb root will alternate depending on the tense/aspect of the sentence. In this section, we provide an analysis of these data. Using Optimality Theory (Prince & Smolensky 1993), we show that there is a difference in the behavior of floating H and floating L grammatical tones. We argue that this difference comes from a need to epenthesize an additional mora to realize the floating grammatical H tone. On the other hand, floating L tones can freely override the underlying tone of the verb root. This difference in behavior has interesting consequences for Igbo tonal phonology and theories of grammatical tone more generally.

We begin this section by providing an overview of the morphosyntax of Igbo verbal grammatical tone (Section 4.1). In this subsection, we argue for specific morphemes in the Igbo verbal complex which are expounded as floating grammatical tones, segmental content, or both. Using Distributed Morphology (Halle & Marantz 1993; 1994; Bobaljik 2017), we tie the tonal alternations observed in the previous section (Section 3) to specific morphosyntactic features, thus showing the precise grammatical function of tone with Igbo verbs. After establishing precisely what morphemes are realized as grammatical tone, we move to a phonological analysis of the tonal alternations (Section 4.2).

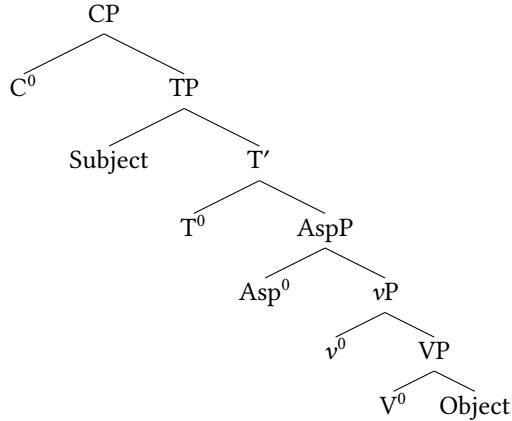
4.1 The Morphosyntax of Grammatical Tone in Igbo Verbs

In Section 3, we laid out the TAM data for Igbo. Now, to provide an analysis of the grammatical tone in these different TAM, it is imperative that we identify

what the different morphemes in the verbal complex are and how they interact with tone.

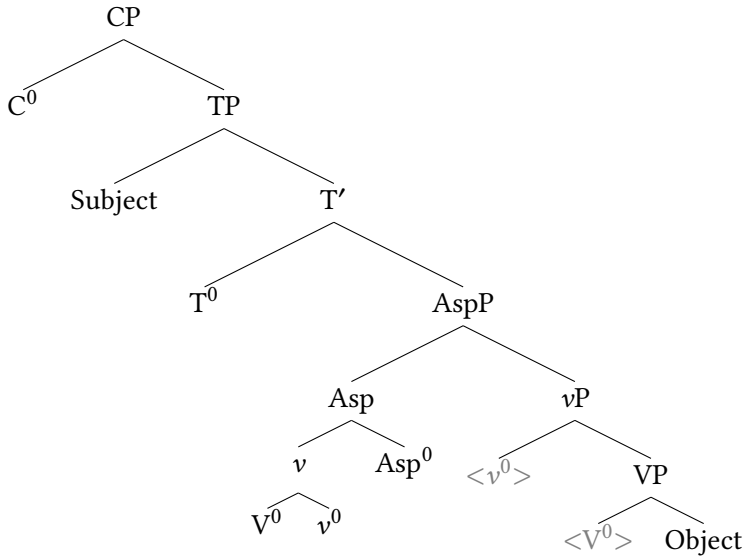
As for the syntactic structure of Igbo verbs, we assume the structure given by Amaechi & Georgi (2019) and Amaechi (2020). This structure is outlined in (11).

(11)



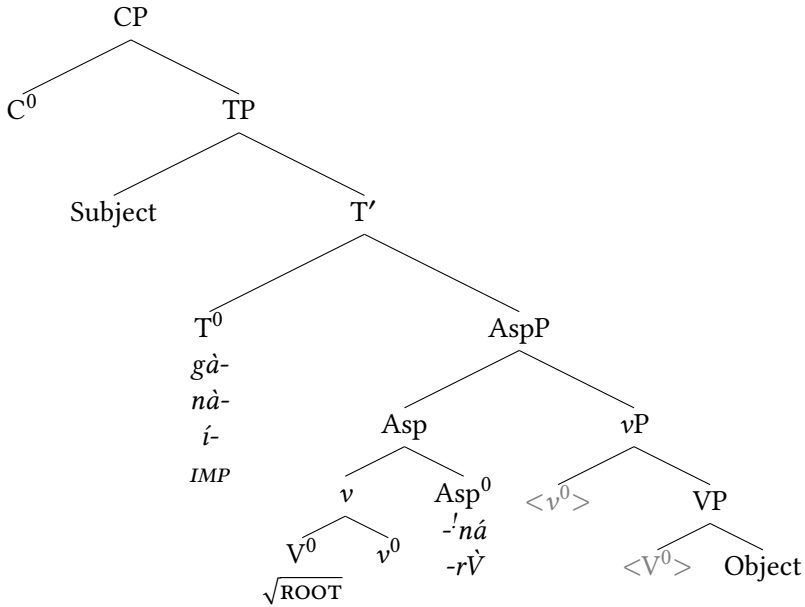
We assume the verb undergoes head-movement to Asp⁰, as in (12).

(12)



Following standard assumptions (Chomsky 1995; 2000), we assume that the infinitive marker /*i*-/ is realization of T^0 . In addition, we assume that aspectual markers, like the perfective marker /-*rV̇*/ and the completive marker /-^H*n'á*/, are the realization of Asp^0 . Furthermore, we take /*gà*-/, the future marker, and /*nà*-/, the progressive marker, to be the realization of T^0 as well. And finally, the imperative, being a morpheme that expresses mood, is also merged in T^0 .³ Thus, the various TAM morphemes are introduced into the syntactic structure as in (13). Explanations for the position of the various TAM morphemes is provided below in the following paragraphs.

(13)



We would like to draw attention to the fact that we place the progressive marker, /*nà*-/, in T^0 . This may seem to be a non-standard assumption as the progressive is usually considered aspectual rather than tense. However, the progressive in Igbo is restricted to the present tense only. In languages where the progressive has aspectual use, it is common to see progressives in both the present and past tenses.⁴ We argue that since the progressive is tense-restricted

³We leave open the possibility that the imperative morphology is introduced in a higher left-peripheral projection like MoodP or ForceP. The precise label of this category is not terribly important at the present time.

⁴For example, in Spanish, *estoy corriendo* is the present progressive for the sentence ‘I am run-

in Igbo, it suggests that we are dealing with a tense marker as opposed to an aspectual one (as we would expect an aspectual marker to be compatible with different tense values). We suggest that the progressive interpretation of the /*nà-*/ morpheme comes from the combination of a T^0 that is marked as present and Asp^0 that is marked as imperfective or incomplete. We suggest that /*nà-*/ is the realization of T^0 in the present tense and that Asp^0 is null. Thus, the semantic interpretation of tense and aspect together leads to progressive semantics, yet the morpheme /*nà-*/ itself is a tense marker.

The proposed structure and placement of the different verbal morphemes provide a solid explanation for the preverbal morpheme-suffix asymmetry. Since head-movement results in left-adjunction (Kayne 1994), the fact that the verb raises to Asp^0 accounts for the suffixal status of the aspect markers. On the other hand, since T^0 c-commands the final landing site of the verb, it is clear why these morphemes appear as preverbal morphemes.

With respect to the verbal morphemes themselves, we take the fact that the infinitive is formed by prefixation of /*i-*/ to the verb root with no change in the underlying tone of the root to be evidence that the infinitive marker is simply the segment /*i-*/. Next, since both /*nà-*/ and /*gà-*/ leave the underlying tone of the verb root unchanged, we believe these are the sole realization of the present and future markers respectively.

The three remaining TAM morphemes always co-occur with a change in the underlying tone of the verb root. Thus, we argue that these TAM morphemes are associated with grammatical tone. First, consider the imperative. In the imperative, verbs, regardless of their underlying tone (H or L), always have a LH tone melody (Clark 1990; Ikegwuonu 2014). Furthermore, in these circumstances, a second TBU is introduced to host the H tone of the LH melody. We will discuss this additional TBU in the next section, but for now, it suffices to say that we consider this TBU to be phonologically conditioned (i.e., it is epenthesized) and not morphologically conditioned (there is no morphological reason for the presence of this extra TBU). We, therefore, propose that the LH melody itself is the realization of the imperative morpheme.

Next, consider the perfective. In the perfective, in addition to the addition of the /-*rV̌*/ suffix, the underlying tone of the verb root is always changed to L. For verbs with an underlying L tone, this change is not surface-noticeable. However,

ning'. It is also possible for the progressive aspect to appear in the imperfect tense (e.g., *estaba corriendo* 'I was running'). Thus, in Spanish, where the progressive is aspectual, it can appear in various tenses (unlike in Igbo where it only occurs in the present tense).

it is clear from the fact that H tone verbs become L in the perfective that there is a tonal overriding in the perfective aspect. We, thus, propose that the perfective morpheme is exponed by $/-^Lr\dot{V}/$. When this suffix combines with the verb root, the floating grammatical L tone docks onto the root and overrides its underlying tone.⁵

Finally, there is the completive. Similar to the imperative, the completive introduces a new TBU to host an extra H tone before the completive suffix. Unlike the perfective, the underlying tone of the verb root remains in the completive, but there is the addition of a new TBU with a H tone. To explain this H tone, we argue that, similar to the perfective, there is a floating grammatical tone at the left edge of the completive morpheme. We propose $/-^Hn'\acute{a}/$ for the completive. In the next section, we go into more depth, but our central claim is that this floating H tone requires realization in the phonology and that TBU epenthesis best captures this pattern. Further evidence for this analysis of the completive comes from downstep in Igbo. Some have argued that a sequence of two H tones (HH) provides the environment for downstep in Igbo (Clark 1990). In these contexts, the second H becomes downstepped (!H). Our investigation led us to the realization that the H tone on the suffix $/-n\acute{a}/$ is, in fact, downstepped ($/-n'\acute{a}/$). Thus, an underlying representation of the completive must have two H tones. The proposed Vocabulary Item for the completive accounts for these facts.

And so, (14) gives the list of TAM morphemes we assume for the Igbo verbal complex.

- (14) a. $[T^0, \text{PRES}] \longleftrightarrow /n\grave{a}/$
 b. $[T^0, \text{FUT}] \longleftrightarrow /g\grave{a}/$
 c. $[T^0, \text{INF}] \longleftrightarrow /i/$
 d. $[T^0, \text{IMP}] \longleftrightarrow /^{LH}/$
 e. $[\text{ASP}^0, \text{PRFV}] \longleftrightarrow /-^Lr\dot{V}/$
 f. $[\text{ASP}^0, \text{COMPL}] \longleftrightarrow /-^Hn'\acute{a}/$
 g. $[\text{ASP}^0, \text{PROG}] \longleftrightarrow /-\emptyset/$

The next section provides a phonological analysis of the grammatical tone alternations in Igbo, assuming the Vocabulary Items introduced in (14) above.

⁵In Sections 5 and 6, we briefly discuss the possibility that this perfective morpheme (and the completive morpheme for that matter) is decomposable. Given the status of this project at the moment, it is not possible to fully consider this possibility here.

4.2 The Phonology of Grammatical Tone in Igbo Verbs

Having now established the precise phonological representations of the TAM morphemes in Igbo, it is possible to analyze the phonological effects the grammatical tones in different TAM combinations. To begin, it is useful to outline some foundational phonological constraints that are pertinent to the current analysis.

Since grammatical tone in Igbo verbs in some cases overrides the underlying tone on the verb root, it is necessary to have an IDENTITY constraint (Prince & Smolensky 1993; McCarthy & Prince 1995) against changing underlying tones (15).

- (15) IDENT(TONE): *Assign one violation for every input tone which has a correspondent in the output that is of a different tonal value (no overriding of underlying tones).*

Additionally, since many of the grammatical tone alternations involve floating tones, a constraint ensuring all tones are linked to a TBU is necessary (16).

- (16) PARSE(TONE): *Assign one violation for every tone in the input which is not linked to a TBU in the output (Yip 2002).*

In general, Igbo does not permit a single TBU to be linked to more than one tone (Ugorji 2008). That is to say, a TBU cannot host a contour tone. Since the TBU in Igbo is the mora, the result is a language-general constraint against having contour tones on a single mora (17).

- (17) *CONTOUR_μ: *Assign one violation for every mora that is associated with more than one tone (Yip 2002).*

Finally, monosyllabic verb roots in Igbo alternate between CV and CVV depending on tense and aspect. As discussed in the previous section (Section 3), we assume that monosyllabic verb roots have an underlying CV structure. The additional V in some tenses/aspects we take to be the result of mora epenthesis. Given this assumption, we need a constraint against epenthesizing additional morae (18).

- (18) DEP(μ): *Assign a violation for every mora in the output that is not present in the input (Prince & Smolensky 1993; McCarthy & Prince 1995).*

These four constraints, ranked as in (19), capture the patterns for the grammatical tone alternations in the imperative, completive, and past tense.

- (19) *CONTOUR_μ, PARSE(TONE) >> IDENT(TONE) >> DEP(μ)

Consider the imperative data in (10), repeated below as (20). For both the L tone root *tò* and the H tone root *gbú* the imperative is realized with a CVV syllable with a LH tone melody.

- (20) a. *tòó àdá*
praise Ada
'praise Ada!'
- b. *gbùó àgù*
kill lion
kill the lion!'

In the previous subsection (Section 4.1), we argued that LH is the realization of the imperative marking in Igbo. At that time, we set aside the issue of the vowel length change. Before demonstrating the OT analysis, let us revisit the issue of vowel length in these grammatical tone constructions. The first thing to notice is that for non-high vowels, the result of the change is to lengthen the vowel to achieve the CVV syllable structure (21).

- (21) a. *tàá (tà)*
chew.IMP
'chew!'
- b. *bèé (bè)*
fly.IMP
'fly!'
- c. *wòó (wò)*
jump.IMP
'jump!'
- d. *màá (má)*
throw.IMP
'throw!'
- e. *dèé (dé)*
write.IMP
'write!'
- f. *gòó (gó)*
deny.IMP
'deny!'

However, when the verb root vowel is a high vowel, there is a lowering of the second vowel in the VV sequence. Thus, high vowels lengthen to a high-mid vowel diphthong (22).

- | | | | |
|------|--------|----------|---------|
| (22) | a. kòs | b. pìá | c. sìé |
| | hit | press | cook |
| | ‘hit!’ | ‘press!’ | ‘cook!’ |

This remains true for multisyllabic verbs as well (23).

- (23) mètʃié ózò
 go.IMP door
 ‘close the door!’

Given this data, it appears that long high vowels in Igbo are not permitted. Thus, to avoid this, Igbo diphthongizes long high vowels. In combination with the fact that non-high vowels simply lengthen, we take this data to be evidence that there is mora epenthesis in the imperative in order to accommodate the LH melody. The vowel quality change can then be added to the OT analysis with the following constraints and ranking (24)-(27).

- (24) *VV([HIGH]): Assign a violation for every long high vowel in the output (adapted from Anttila & Bodomo (2009) for mid vowels).
- (25) IDENT([HIGH]): Assign one violation for every vowel in the input which has a correspondent in the output that has a different value for the [HIGH] feature (Prince & Smolensky 1993; McCarthy & Prince 1995).
- (26) IDENT([LOW]): Assign one violation for every vowel in the input which has a correspondent in the output that has a different value for the [LOW] feature (Prince & Smolensky 1993; McCarthy & Prince 1995).
- (27) *VV([HIGH]) >> IDENT([LOW]) >> IDENT([HIGH])

To demonstrate how these constraints and the rankings above account for the data, consider the tableaux below. The tableau in (28) shows a derivation for the imperative of the H tone verb *gbú* ‘kill’. *gbú*, being a verb with a high vowel, undergoes diphthongization in the imperative. The tableau in (29) provides a derivation for the imperative of the L tone verb *tò* ‘praise’. This verb root has a mid vowel, and as a result, no diphthongization takes place.

(28)

LH- $\widehat{gbú}$	*VV([HIGH])	PARSE(TONE)	*CONT $_{\mu}$	IDENT([LOW])	IDENT([HIGH])	IDENT(TONE)	DEP(μ)
 a. $\widehat{gbù}ó$					*	*	*
b. $\widehat{gbù}á$				*!	*	*	*
c. $\widehat{gbù}$			*!			*	
d. $\widehat{gbù}$		*!				*	
e. $\widehat{gbù}$		*!					
f. $\widehat{gbù}ú$	*!					*	*

(29)

LH-tò	*VV([HIGH])	PARSE(TONE)	*CONT $_{\mu}$	IDENT([LOW])	IDENT([HIGH])	IDENT(TONE)	DEP(μ)
 a. tòó							*
b. tòú					*!		*
c. tòá				*!			*
d. tò			*!			*	
e. tò		*!					
f. tò		*!				*	

This same ranking accounts for the completive as well. Recall that the completive is expressed by the morpheme $Hn'á$. The tableau in (30) and (31) provide derivations of the completive for $\widehat{gbù}$ and $tò$, respectively.

(30)

$\widehat{gbù}$ - $Hn'á$	*VV([HIGH])	PARSE(TONE)	*CONT $_{\mu}$	IDENT([LOW])	IDENT([HIGH])	IDENT(TONE)	DEP(μ)
 a. $\widehat{gbù}ón'á$					*		*
b. $\widehat{gbù}án'á$				*!	*		*
c. $\widehat{gbùn}á$		*!					
d. $\widehat{gbù}ún'á$	*!						*

(31)

tò- $Hn'á$	*VV([HIGH])	PARSE(TONE)	*CONT $_{\mu}$	IDENT([LOW])	IDENT([HIGH])	IDENT(TONE)	DEP(μ)
 a. tòón'á							*
b. tôn'á						*!	
c. tòún'á					*!		*
d. tòán'á				*!			*
e. tôn'á			*!			*	
f. tòn'á		*!					

Finally, (32) and (33) provide derivations for the perfective.

(32)

$\widehat{gb\acute{u}}-{}^Lr\check{V}$	*VV([HIGH])	PARSE(TONE)	*CONT _{μ}	IDENT([LOW])	IDENT([HIGH])	IDENT(TONE)	DEP(μ)
a. $\widehat{gb\acute{u}}r\grave{u}$						*	
b. $\widehat{gb\acute{u}}\grave{o}r\grave{u}$					*!		*
c. $\widehat{gb\acute{u}}\grave{a}r\grave{u}$				*!	*		*
d. $\widehat{gb\acute{u}}r\grave{u}$			*!			*	
e. $\widehat{gb\acute{u}}r\grave{u}$		*!					
f. $\widehat{gb\acute{u}}\grave{u}r\grave{u}$	*!						*

(33)

$t\grave{o}-{}^Lr\check{V}$	*VV([HIGH])	PARSE(TONE)	*CONT _{μ}	IDENT([LOW])	IDENT([HIGH])	IDENT(TONE)	DEP(μ)
a. $t\grave{o}r\grave{o}$							
b. $t\grave{o}\grave{o}r\grave{o}$							*!
c. $t\grave{o}\grave{u}r\grave{o}$					*!		*
d. $t\grave{o}\grave{a}r\grave{o}$				*!			*

As should be clear from the tableaux above, the ranking provided in (34) correctly accounts for the tonal alternations for all of the TAM combinations that have grammatical tone in Igbo. This ranking also has the benefit of accounting for the vowel length changes exhibited by the root depending on whether the vowel tone is a high vowel or not. Finally, these derivations demonstrate how floating grammatical H tone requires the epenthesis of an additional mora, whereas floating grammatical L tone overrides the underlying tone of the verb root.

(34) *VV([HIGH]), PARSE(TONE), *CONT _{μ} >> IDENT([LOW]) >> IDENT([HIGH]) >> IDENT(TONE) >> DEP(μ)

In the next section (Section 5), we discuss these findings in greater detail and examine the implications of our findings.

5 Discussion and Implications

In this section, we discuss the findings of our analysis and consider the implications of these findings for linguistic theory. We begin by discussing our findings and examining what these findings tell us about Igbo morphosyntax, phonology, and tone. Then, we move to investigating the greater implications of these findings.

5.1 Empirical Findings

Tonal overriding in Igbo verbal morphology refers to the systematic replacement of a verb root's underlying lexical tone by grammatically specified tones. This phenomenon is particularly evident in the perfective and imperative constructions. In the perfective aspect, the verb root is suffixed with the $/^{-L}r\check{V}/$ morpheme, where the vowel (V) is a copy of the root-final vowel and obligatorily bears a L tone. This L tone suffix triggers overrides any underlying tone on the verb stem.

In the imperative construction, tonal overriding also occurs, but with a distinct $/^{LH}/$ tone melody (Clark 1990; Ikegwuonu 2014). The initial low tone replaces a verb root's underlying tone, while the following high tone surfaces on an additional TBU through vowel lengthening. This tonal restructuring ensures that the grammatical tone sequence is properly realized and does not create a contour tone on a single TBU (Clark 1990; Ugorji 2008).

Our analysis further reveals that tonal overriding is absent in progressive, future, completive, and infinitive constructions in Igbo. In these contexts, the lexical tone of the verb stem is preserved, and tone is instead inherent to functional TAM morphemes, such as preverbal morphemes $/n\grave{a}-/$ (progressive), $/g\grave{a}-/$ (future), or infinitival prefixes $/i-/i-/$.

Uniquely, in the completive, the lexical tone of the verb root remains preserved, and the floating grammatical H tone is realized on an epenthetic mora, rather than overriding the stem's tonal specification. This indicates that grammatical tone in these constructions is additive rather than overriding, and this aligns with principles of tonal preservation and prosodic accommodation (Akinlabi & Liberman 2007).

In this way, the completive and imperative constructions illustrate the clearest interaction between segmental phonology and tonal grammar, with the insertion of an epenthetic mora to realize the floating grammatical H tone. This serves as a functional strategy not only for accommodating tonal sequences but also for preventing the occurrence of unassociated floating tones in the output.

These tonal alternations and the subsequent analysis support strong evidence for a syntax-prosody interface in Igbo where morphological features are not only realized through affixation but also through systematic tonal operations like overriding by floating grammatical L tones and mora epenthesis for H tones.

Thus, we see an asymmetrical behavior of grammatical tones in Igbo. Specifically, grammatical L tone can override a verb root's underlying tone, and grammatical H tone forces mora epenthesis to provide a TBU for the tone's realization. This is an interesting finding as it highlights how tone behaves in different phonological ways despite similar morphophonological environments. However, when considering the behavior of tone in Igbo more generally, it is not entirely surprising that the grammatical tones behave in this way.

Clark (1990) argues that H is the default tone in Igbo. As such, tonal overriding by H would constitute a neutralization to the language's default tone value. Although this would constitute an example of the Emergence of the Unmarked, such a phonological neutralization would obscure the morphological changes that verbs are undergoing in these grammatical tone contexts. Put differently, the fact that L is not the default tone in Igbo provides some explanation for the different behavior of grammatical H and L tones. Grammatical H tones receive an epenthetic mora to indicate that the tone is morphological and not a phonological neutralization. On the other hand, L tone is not default or neutralized in the language, so tonal overriding by L necessarily indicates a morphological change since it is a marked change in the phonology.

In this way, we see that Igbo has two strategies for realizing grammatical tone. When the grammatical tone is a marked tone (L), overriding on the root is possible because the marked change is a salient morphological change. However, when the grammatical tone is the unmarked tone in the language (H), additional changes must also occur in order to maintain the contrast between the morphosyntactically important H tone and neutralization of the H tone in the phonology. This relationship between phonological markedness and morphosyntactic expression provides an interesting area of exploration in grammatical tone and nonconcatenative morphology more generally.

5.2 Theoretical Consequences and Implications

The findings in this paper also have far-reaching theoretical implications for linguistics research. (35) provides a nonexhaustive list of some of such implications.

(35)

(i) Grammatical Tone as Morphological Exponence:

The data offer robust evidence that tone functions as a morphological exponent in Igbo, marking tense, aspect, and mood (TAM) features without segmental affixation. This challenges traditional, segmentally focused morphology models and supports theories in which suprasegmental features, especially tone, serve as fully grammatical morphemes encoding aspectual contrast.

(ii) Asymmetry in the Behavior of Grammatical L and H Tones:

A key theoretical insight emerging from this study is the asymmetrical behavior of grammatical L and H tones. Grammatical L tones can override underlying lexical tone (e.g., in the perfective), while grammatical H tones require prosodic augmentation (e.g., mora epenthesis) for realization. This asymmetry is rooted in markedness theory, where H is conceived as the default/unmarked tone in Igbo. Therefore, overriding a root with an H tone would result in neutralization, obscuring the morphological distinction. To avoid this, Igbo realizes grammatical H tones on new TBUs, maintaining a phonological contrast between lexical H and grammatical H. In contrast, L is marked and cannot be phonologically neutralized, so its overwriting effect is both phonologically salient and morphosyntactically interpretable.

(iii) Constraint Interaction and Vowel Quality Effects:

The need to avoid long high vowels (as a result of the *VV[HIGH] constraint) in imperative and completive constructions further shows how segmental constraints interact with tonal morphology. Vowel lowering or diphthongization arises to satisfy markedness constraints on vowel height while preserving TBUs for grammatical tones. This reinforces the view that ATR harmony, vowel height, and tone are co-regulated by a hierarchy of constraints (e.g., *VV[HIGH] » IDENT[LOW] » IDENT[HIGH]). It also highlights that segmental and tonal grammar must be analyzed as part of an integrated system in Igbo grammar.

(iv) Implications for Theories of Nonconcatenative Morphology:

These findings contribute to the broader typology of nonconcatenative morphology, where morphological features are realized through phonological change operations rather than segmental affixation. Igbo demonstrates two distinct strategies for realizing grammatical tone—(1) tonal overriding when the tone is phonologically marked (i.e., L) and (2) prosodic mora insertion when the tone is unmarked or default (i.e., H). This supports

the view that phonological markedness interacts with morphological exponence, shaping how languages distinguish between grammatical and phonological tone.

(v) **Typological Implications:**

From a typological and cross-linguistic perspective, Igbo exemplifies a grammatical tone system in which tone plays a central role in clause structure, serving lexical contrast and functioning as a primary marker of grammatical information such as tense, aspect, and mood.

6 Conclusion and Future Directions

This study has shown that grammatical tone in Igbo verbs operates as a key morphosyntactic exponent for encoding tense, aspect, and mood. Beyond its lexical function, tone in Igbo plays an active grammatical role, systematically interacting with morphosyntactic structure, vowel length, and prosodic well-formedness constraints. While tonal overriding characterizes constructions like the perfective and imperative, where grammatical tone replaces the root's underlying tone, constructions such as the progressive, future, infinitive, and completive preserve the verb's lexical tone, assigning grammatical tone to functional morphemes or epenthetic morae.

This study demonstrated that the tonal and segmental behavior of these constructions can be formally analyzed within Optimality Theory and Distributed Morphology, where the phonological exponents of different Vocabulary Items are evaluated against a ranked constraint grammar interacting in various ways to generate the observed outputs. Constraints such as *PARSE(TONE)* and **CONTOUR_μ* ensure tonal association and prohibit contour tones on a single mora, while *IDENT(TONE)* and *DEP(μ)* govern tone faithfulness and mora epenthesis. The ranking **CONTOUR_μ*, *PARSE(TONE)* » *IDENT(TONE)* » *DEP(μ)* captures the grammar's preference for realizing grammatical tone over preserving underlying tone or, in some cases, triggering mora insertion.

A more complex interaction emerges in the imperative construction, where mora epenthesis is employed to host a /^{LH}/ grammatical tone melody. In cases where the root ends in a high vowel, the language avoids ill-formed long high vowels by lowering the second vocalic mora to a mid vowel. The ranking **VV([HIGH])* » *IDENT([LOW])* » *IDENT([HIGH])* accounts for the asymmetry whereby non-high vowels undergo lengthening while long high vowels shift in quality to satisfy

both tonal and segmental well-formedness.

Building on the present analysis, future work will extend the investigation to a broader range of Igbo verb structures, including disyllabic, trisyllabic, and compound verbs, to determine whether the patterns of grammatical tone observed in monosyllabic verbs generalize across more complex verbal forms.

Additionally, we aim to explore the interaction between grammatical tone and syntactic structure at the higher-level phonology-syntax interface, focusing in particular on tonal behavior in relative clause and focus constructions in Igbo as well as the so-called “N-N Associative Construction”. Preliminary observations suggest that in certain subject relative clauses, a grammatical high tone (H) docks onto the final vowel of the subject noun phrase and spreads onto subsequent verbal elements (Amaechi 2020). Notably, this tonal spreading is absent in object relative clauses and focus constructions, where the grammatical H tone does not propagate beyond the subject. Investigating these asymmetries will provide further insight into how syntactic configurations condition tonal realization and will contribute to a deeper understanding of tone-driven morphosyntax in complex syntactic environments.

Finally, we will consider the possibility that the floating grammatical tones in the completive and perfective aspects are, in fact, separate morphemes from the segmental content in these suffixes. Clark (1990) suggests that the floating H tone in the completive might be the realization of affirmative polarity. This is entirely possible, but without investigation into the grammatical tone patterns of negation in Igbo, such a position cannot be confirmed at present.

To conclude, altogether, our study provides strong evidence that grammatical tone in Igbo verbs is governed by a rich interaction of morphosyntactic features, tonal grammar, and segmental phonology. The findings reinforce the central role of tone in the inflectional system and highlight the relevance of constraint-based phonological theory in explaining both tonal overriding and vowel quality alternations in the same grammatical tone environments.

Abbreviations

In this paper, we follow the Leipzig Glossing Conventions with the following additions: *v*, which represents the initial vowel that surfaces on verbs in some tense-aspect-mood paradigms. At present, we do not wish to take a stance as to what this morpheme is since it does not have an impact on our analysis.

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References

- Akinlabi, A. & M. Liberman. 2007. Tonal complexes: the prosodic organization of tones. *Rutgers Optimal Archive* 911.
- Amaechi, M. 2020. *A'-movement dependencies and their reflexes in igbo*. University of Potsdam. (Doctoral dissertation).
- Amaechi, M. & D. Georgi. 2019. Quirks of subject (non)-extraction in igbo. *Glossa* 4(1). 1–36.
- Anttila, A. & A. Bodomo. 2009. Prosodic morphology in dagaare. In *Selected proceedings of the 38th annual conference on african linguistics*. Cascadilla Proceedings Project.
- Armstrong, R.C. 1967. *A comparative wordlist of five igbo dialects*. Institute of African Studies.
- Bobaljik, J. 2017. Distributed morphology. In *Oxford research encyclopedia of linguistics*. Oxford University Press.
- Chomsky, N. 1995. *The minimalist program*. Minimalism. The MIT Press.
- Chomsky, N. 2000. Minimalist inquiries: the framework. In *Step by step: essays in minimalist syntax in honor of howard lasnik*, 89–155. MIT Press.
- Clark, M. 1990. *The tonal system of igbo*. de Gruyter.
- Emenanjo, E. 1978. *Elements of modern igbo grammar*. Oxford University Press.
- Emenanjo, E. 1985. *Auxiliaries in igbo syntax: a comparative study*. IULC.
- Emenanjo, E. 2016. *A grammar of contemporary igbo: constituents, features and processes*. M & J Grand Orbit Communications.
- Halle, M. & A. Marantz. 1993. Distributed morphology and the pieces of inflection. In Kenneth Hale & S. Jay Keyser (eds.), *The view from building 20*, 111–176. MIT Press.
- Halle, M. & A. Marantz. 1994. Some key features of distributed morphology. In *MIT working papers in linguistics*, vol. 21, 275–288. MIT Press.

- Ikegwonu, C. 2014. Expression of imperative mood in igbo. *Journal of Linguistics, Language, and Culture* 1(2).
- Ikekeonwu, C. 1999. Igbo. In *Handbook of the international phonetic association*. Cambridge University Press.
- Kayne, R. 1994. *The antisymmetry of syntax*. MIT University Press.
- Mbah, B.M. 1999. *Studies in syntax: igbo phrase structure*. Nsukka: Prize Publishers.
- McCarthy, J. & A. Prince. 1995. Faithfulness and reduplicative identity. *University of Massachusetts Occasional Papers in Linguistics* 18.
- Nkamigbo, L. 2014. Nasalization, aspiration, and labialization in igbo dialect phonology. *Skase Journal of Theoretical Linguistics* 11. 59–67.
- Nweke, J. 2012. Igbo vowel harmony: segmental and autosegmental approaches. *Multidisciplinary Journal of Academic Excellence*.
- Obiamalu, G.O. 2016. *Functional categories in igbo: a minimalist perspective*. Vol. 4. M & J Grand Orbit Communications.
- Prince, A. & P. Smolensky. 1993. *Optimality theory: constraint interactions in generative grammar*. Rutgers University and The Johns Hopkins University.
- Ugorji, C.U.C. 2008. Contour tones in igbo. *Language in India* 8. 1–16.
- Yip, M. 2002. *Tone* (Cambridge Textbooks in Linguistics). Cambridge, UK: Cambridge University Press.
- Zsiga, E. 1992. A mismatch between morphological and prosodic domains: evidence from two igbo rules. *Phonology* 9. 101–135.